

Shruti Debnath

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EDUCATION

University of Idaho | *Recipient of the Invitation to Idaho Scholarship*

Idaho, USA

Bachelor of Science in Computer Science | AI/ML Focus

December 2026

GPA: 3.48/4.0 | Dean's List (3 semesters)

Relevant Coursework: Data Structures, Algorithms, Database Systems, Software Engineering, Machine Learning

TECHNICAL SKILLS

- **Languages:** Python, TypeScript, JavaScript, C#, C, C++
- **Frameworks & Libraries:** React, Next.js, Node.js, Django, Tailwind CSS, Framer Motion, REST APIs
- **Databases:** PostgreSQL, MySQL, SQLite
- **Tools:** Git, Unity, Render, Vercel
- **Data/ML libraries:** PyTorch, Scikit-learn, NumPy, Pandas

PROJECTS

Interactive Algorithm Visualizer

TypeScript, React, Vite, Tailwind CSS, Vercel

- Developed an interactive algorithm visualizer to simulate sorting algorithms with real-time, step-by-step execution.
- Implemented visualization for algorithms including Bubble Sort, Merge Sort, BFS, and DFS, enabling users to observe algorithm behavior and time complexity.
- Designed dynamic UI components to control execution speed, input size, and algorithm selection, improving user interaction and learning experience.

CATScan — Automated Security Testing Tool for C/C++

Python, Regex, LibClang, C/C++, HTML

- Built an automated tool to perform static and dynamic security analysis on C/C++ programs.
- Implemented testcase generation for vulnerabilities including buffer overflow, integer overflow, divide-by-zero, blank input, and format string issues.
- Automated compilation, execution, timeout handling, and HTML report generation to summarize runtime failures and detected vulnerabilities.

Barter App

Flask, REST APIs, Python, SQLite, HTML, Git

- Developed a full-stack barter platform using Flask and SQLite, designing a 6-table relational schema for users, items, trades, partnerships, and transactions.
- Implemented 26+ REST API endpoints for authentication, item listing, trade proposals, profile management, and admin operations.
- Engineered a two-step trade verification system using split 16-digit codes, ensuring secure and mutual transaction confirmation.

Combat Game

Unity, C#, Git

- Led a team of 7 developers to architect a modular 2D combat game in Unity, implementing object-oriented design and dynamic binding across gameplay and UI systems.
- Designed and managed 8+ interactive UI panels with Apply/Discard state management, centralized configuration handling, and controlled scene transitions.
- Developed and executed 30+ unit and boundary tests and 5 stress tests to validate gameplay logic and UI workflows, while organizing version control using structured Git workflows.

WORK EXPERIENCE

SI-PASS Leader

Aug 2024 – Present

University of Idaho — College Success (Tutoring)

Idaho, USA

- Led weekly SI sessions for ~20 students, increasing participation and improving exam performance in Pre-Calculus.
- Contributed to ~10% improvement in average midterm scores among consistent attendees.
- Collaborated with course instructor to monitor discussions and resolve conceptual gaps in real time.

Lead Tutor

Oct 2025 – Present

University of Idaho — Math Assistance Center (MAC)

Idaho, USA

- Led bi-weekly strategy meetings with 14 tutors to align coursework updates and improve instructional consistency.
- Collaborated with supervisors to implement engagement-focused initiatives, increasing student attendance.
- Provided one-on-one and drop-in tutoring support to improve student performance in mathematics courses.
- Developed training materials aligned with CRLA certification standards.